

MAT1575, Classwork1, Fall2026

4.10 Antiderivatives (p.419-429)

1. Find the derivative of the given function.

- (a) $F(x) = x^3 + x^2 + 1$ (b) $F(x) = \ln|x|$ (c) $F(x) = e^x$ (d) $F(x) = \cos(x)$ (e) $F(x) = x^3 + x^2 - 20$

(a) $F'(x) = 3x^2 + 2x$ (b) $F'(x) = \frac{1}{x}$ (c) $F'(x) = e^x$ (d) $F'(x) = -\sin(x)$ (e) $F'(x) = 3x^2 + 2x$

$\frac{dF(x)}{dx}$

2. Definition: What is an Antiderivative of a given function?

A function F is an antiderivative of the function f if, for all x in the domain of f ,

$F'(x) = f(x)$

$\frac{dF(x)}{dx} = f(x)$

Observation: The antiderivative of the function f is a function with a derivative. (The reverse of differentiation)

3. Applications: Why would we be interested in an Antiderivative of a given function?

$\frac{ds(t)}{dt}$

$v(t) = s'(t)$

$a(t) = v'(t)$

Physics(rectilinear motion): position function $s(t)$ velocity function $v(t)$ acceleration function $a(t)$

4. General Form of An Antiderivative.

Let $F(x)$ be an antiderivative of the function $f(x)$ over an interval I . Then,

$\frac{dF(x)}{dx} = f(x)$

(a) $F(x) + C$ is also an antiderivative of the function $f(x)$ over I for any arbitrary constant C .

$\frac{d}{dx}(F(x) + C)$

(b) The most general form of the antiderivative of $f(x)$ over I is $F(x) + C$.

$= \frac{d}{dx} F(x) + \frac{d}{dx} C$

5. For each of the following functions, find all antiderivatives.

(a) $f(x) = 3x^2 + 2x$

(b) $f(x) = \frac{1}{x}$

(c) $f(x) = e^x$

(d) $f(x) = \cos(x)$

(a) $F(x) = \frac{3x^{2+1}}{3} + \frac{2x^{1+1}}{2} = x^3 + x^2 + C$

(b) $F(x) = \ln|x| + C$

(c) $F(x) = e^x + C$
 $F'(x) = e^x$

(d) $F(x) = \sin(x) + C$

6. Definition: The Indefinite Integrals of a function f and the notation of integrations.

Given a function f . The indefinite integral of f is the most general antiderivative of f , denoted

$\int f(x) dx$

where the integral symbol $\int$ represents integration, $f(x)$ is called the integrand, and the symbol dx called the differential of the variable which indicates that variable of integration is x and represents an infinitesimally small change in x .

If function F is an antiderivative of f , then $\int f(x) dx = F(x) + C$

a) function
b) number
c)

7. Integration Formulas

Differentiation Formula	Indefinite Integral	Differentiation Formula	Indefinite Integral
$\frac{d}{dx}(k) = 0$	$\int k dx = k \cdot x + C$	$\frac{d}{dx}(x^n) = n \cdot x^{n-1}$	$\int x^n dx = \frac{x^{n+1}}{n+1}$ for $n \neq -1$
$\frac{d}{dx}(\ln x) = \frac{1}{x}$	$\int \frac{1}{x} dx =$	$\frac{d}{dx}(e^x) = e^x$	$\int e^x dx =$
$\frac{d}{dx}(\sin(x)) = \cos(x)$	$\int \cos(x) dx =$	$\frac{d}{dx}(\cos(x)) = -\sin(x)$	$\int \sin(x) dx =$
$\frac{d}{dx}(\tan(x)) = \sec^2(x)$	$\int \sec^2(x) dx =$	$\frac{d}{dx}(\csc(x)) = -\csc(x) \cot(x)$	$\int \csc(x) \cot(x) dx =$
$\frac{d}{dx}(\cot(x)) = -\csc^2(x)$	$\int \csc^2(x) dx =$	$\frac{d}{dx}(\sec(x)) = \sec(x) \tan(x)$	$\int \sec(x) \tan(x) dx =$
$\frac{d}{dx}(\sin^{-1}(x)) = \frac{1}{\sqrt{1-x^2}}$	$\int \frac{1}{\sqrt{1-x^2}} dx =$	$\frac{d}{dx}(\tan^{-1}(x)) = \frac{1}{1+x^2}$	$\int \frac{1}{1+x^2} dx =$
$\frac{d}{dx}(\sec^{-1} x) = \frac{1}{x\sqrt{x^2-1}}$	$\int \frac{1}{x\sqrt{x^2-1}} dx =$		

8. Properties of Indefinite Integrals.

Let F and G be antiderivatives of f and g respectively, and let k be any real number.

(a) Sums and Differences: $\int (f(x) + g(x)) dx =$
 $\int (f(x) - g(x)) dx =$

(b) Constant Multiples: $\int kf(x) dx =$

9. Evaluate each of the following indefinite integrals:

(a) $\int (5x^3 - 7x^2 + 3x + 4) dx$ (b) $\int \frac{x^2 + 4\sqrt[3]{x}}{x} dx$ (c) $\int \frac{4}{1+x^2} dx$ (d) $\int \tan(x) \cos(x) dx$

MAT1575, Classwork2, Fall2026

1.1 Approximating Areas (p.6-23)

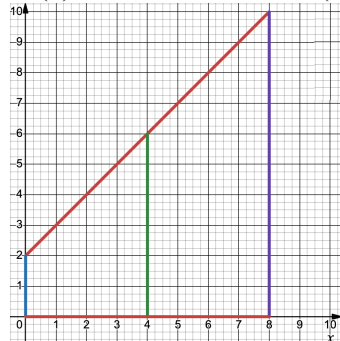
1. Example: Approximating Area of a Trapezoid under a given line.

Given a function $f(x) = x + 2$. Please finish the following questions:

(a) Find the area between $y = 0$ and $f(x)$ for $0 < x < 8$.

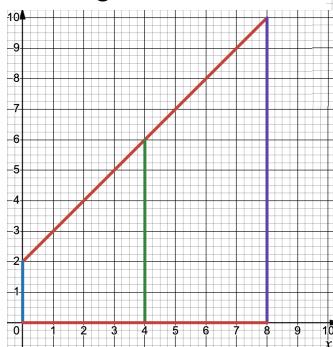
$A =$

(b) Estimate the area in (a) by dividing it into 2 equal-width rectangles:



Left-endpoint approximation

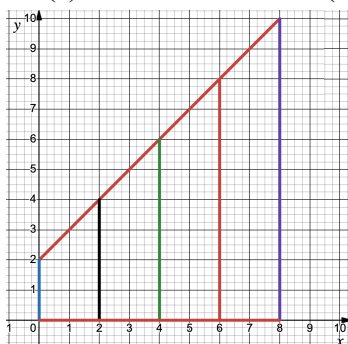
$L_2 =$



Right-endpoint approximation

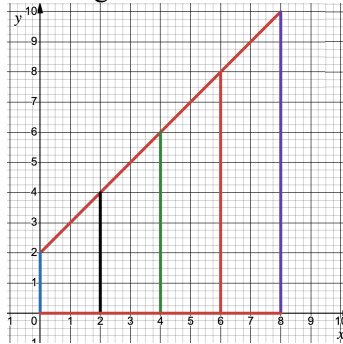
$R_2 =$

(c) Estimate the area in (a) by dividing it into 4 equal-width rectangles:



Left-endpoint approximation

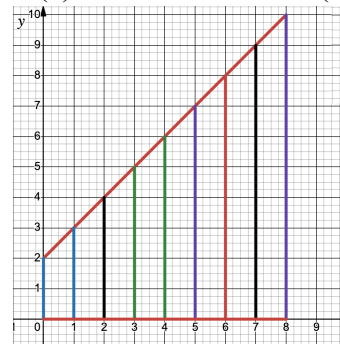
$L_4 =$



Right-endpoint approximation

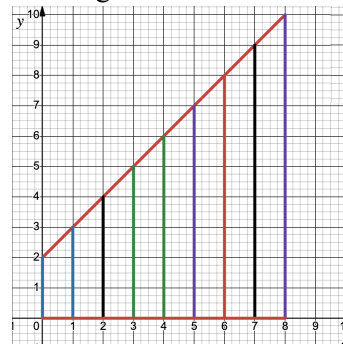
$R_4 =$

(d) Estimate the area in (a) by dividing it into 8 equal-width rectangles:



Left-endpoint approximation

$L_8 =$



Right-endpoint approximation

$R_8 =$

(e) What is the relationship between $A, L_2, R_2, L_4, R_4, L_8$, and R_8 ?

2. Definition: Partition

A set of points $P = \{x_i\}$ for $i = 1, 2, \dots, n$ with $a = x_0 < x_1 < x_2 < \dots < x_n = b$, which divides the interval $[a, b]$ into subintervals of the form $[x_0, x_1], [x_1, x_2], \dots, [x_{n-1}, x_n]$ is called _____.

If the subintervals all have _____, the set of points forms a _____ of the interval.

Example in 1(b): $P =$ _____ it divides $[0, 8]$ into _____ which is a _____ and width $\Delta x =$ _____

Example in 1(c): $P =$ _____ it divides $[0, 8]$ into _____ which is _____ and width $\Delta x =$ _____

3. Left-endpoint approximation L_n and Right-endpoint approximation R_n

On each subinterval $[x_{i-1}, x_i]$ (for $i = 1, 2, \dots, n$), we construct a rectangle with width Δx and height h to calculate the area of this rectangle is _____. Adding the areas of all these rectangles, we get an _____.

(a) Left-endpoint approximation L_n

$h =$ _____ which is the function value at the _____ of the subinterval and

$$L_n =$$

$$=$$

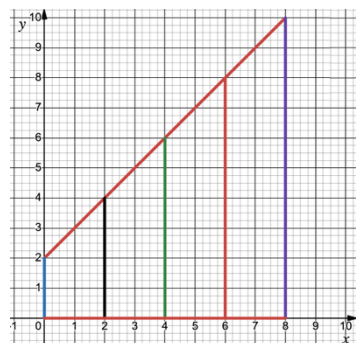
(b) Right-endpoint approximation R_n

$h =$ _____ which is the function value at the _____ of the subinterval and

$$R_n =$$

$$=$$

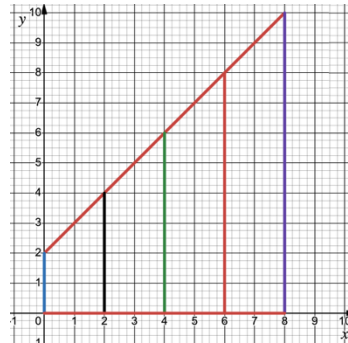
Example: $f(x) = x + 2$ with $P =$ _____



$n =$

$L_n =$

Example: $f(x) = x + 2$ with $P =$ _____



$n =$

$R_n =$

4. Review: Sums and Powers of Integers.

(a) The sum of n integers is given by $\sum_{i=1}^n i =$

(b) The sum of consecutive integers squared is given by $\sum_{i=1}^n i^2 =$

5. Review: Properties of Sigma Notation.

Let $a_1, a_2, \dots, a_n$ and $b_1, b_2, \dots, b_n$ represent two sequences of terms and let c be a constant.

(a) $\sum_{i=1}^n c = nc.$

(b) $\sum_{i=1}^n ca_i = c \sum_{i=1}^n a_i$

(c) $\sum_{i=1}^n (a_i + b_i) = \sum_{i=1}^n a_i + \sum_{i=1}^n b_i$

(d) $\sum_{i=1}^n (a_i - b_i) = \sum_{i=1}^n a_i - \sum_{i=1}^n b_i$

6. Example.

(a) Find the sum of the terms $(i - 3)^2$ for $i = 1, 2, \dots, 200$.

(b) Find the sum of the terms $j(j + 1)$ for $j = 0, 1, \dots, 9$.